A SHEAF PRESENTED BY FINITE DATA ON A SPECTRAL SPACE COMES FROM A FINITE SOBER QUOTIENT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-modules/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The presentation of a spectral space as a directed limit of finite sober spaces, with spectral transition maps and hence spectral projections. Invoked at the start of the counted proof (source lines 3289-3294) as cross-chapter pointers. Provides the finite quotients on which the presentation is to be descended.
- The descent of opens and of maps into a colimit of sheaves along the projections of a directed limit. Invoked inside the counted proof (source lines 3295-3297 and 3316-3319) as cross-chapter pointers. Allows the finitely many opens and the finitely many values of the two maps to be brought down to some stage of the limit.

Source reference. Original equation equation-towards-constructible-sets: sources/stacks-modules/pinned-original-source.tex, source lines 3281–3281. Navigation pointer only: the counted text refers to the Stacks item equation-towards-constructible-sets, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference topology-lemma-spectral-inverse-limit-finite-sober-spaces: sources/stacks-modules/pinned-original-source.tex, source lines 3291–3291. Navigation pointer only: the counted text refers to the Stacks item topology-lemma-spectral-inverse-limit-finite-sober-spaces, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

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Source reference. Original reference topology-lemma-directed-inverse-limit-spectral-spaces: sources/stacks-modules/pinned-original-source.tex, source lines 3293–3293. Navigation pointer only: the counted text refers to the Stacks item topology-lemma-directed-inverse-limit-spectral-spaces, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference topology-lemma-descend-opens: sources/stacks-modules/pinned-original-source.tex, source lines 3297–3297. Navigation pointer only: the counted text refers to the Stacks item topology-lemma-descend-opens, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference sheaves-lemma-compute-pullback-to-limit: sources/stacks-modules/pinned-original-source.tex, source lines 3317–3317. Navigation pointer only: the counted text refers to the Stacks item sheaves-lemma-compute-pullback-to-limit, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let X be a spectral topological space. Let $\mathcal{B}$ be the set of quasi-compact open subsets of X. Let $\mathcal{F}$ be a sheaf of sets as in Equation (equation-towards-constructible-sets). Then there exists a continuous spectral map $f : X \to Y$ to a finite sober topological space Y and a sheaf of sets $\mathcal{G}$ on Y with finite stalks such that $f^{-1}\mathcal{G} \cong \mathcal{F}$.*

Proof. We can write $X = \lim X_i$ as a directed limit of finite sober spaces, see Topology, Lemma topology-lemma-spectral-inverse-limit-finite-sober-spaces. Of course the transition maps $X_{i'} \to X_i$ are spectral and hence by Topology, Lemma topology-lemma-directed-inverse-limit-spectral-spaces the maps $p_i : X \to X_i$ are spectral. For some i we can find opens $U_{a,i}$ and $V_{b,i}$ of X_i whose inverse images are U_a and V_b, see Topology, Lemma topology-lemma-descend-opens. The two maps

$$\beta, \gamma : \coprod\nolimits_{b \in B} j_{V_b!}\underline{S_b} \longrightarrow \coprod\nolimits_{a \in A} j_{U_a!}\underline{S_a}$$

whose coequalizer is $\mathcal{F}$ correspond by adjunction to two families

$$\beta_b, \gamma_b : S_b \longrightarrow \Gamma(V_b, \coprod\nolimits_{a \in A} j_{U_a!}\underline{S_a}), \quad b \in B$$

of maps of sets. Observe that $p_i^{-1}(j_{U_{a,i}!}\underline{S_a}) = j_{U_a!}\underline{S_a}$ and $(X_{i'} \to X_i)^{-1}(j_{U_{a,i}!}\underline{S_a}) = j_{U_{a,i'}!}\underline{S_a}$. It follows from Sheaves, Lemma sheaves-lemma-compute-pullback-to-limit (and using that S_b and B are finite sets) that after increasing i we find maps

$$\beta_{b,i}, \gamma_{b,i} : S_b \longrightarrow \Gamma(V_{b,i}, \coprod\nolimits_{a \in A} j_{U_{a,i}!}\underline{S_a}), \quad b \in B$$

which give rise to the maps β_b and γ_b after pulling back by p_i. These maps correspond in turn to maps of sheaves

$$\beta_i, \gamma_i : \coprod\nolimits_{b \in B} j_{V_{b,i}!}\underline{S_b} \longrightarrow \coprod\nolimits_{a \in A} j_{U_{a,i}!}\underline{S_a}$$

on X_i. Then we can take $Y = X_i$ and

$$\mathcal{G} = \text{Coequalizer}\Big(\coprod\nolimits_{b=1,\ldots,m} j_{V_{b,i}!}\underline{S_b} \rightrightarrows \coprod\nolimits_{a=1,\ldots,n} j_{U_{a,i}!}\underline{S_a} \Big)$$

We omit some details. □